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6.3.R1 (1/1 point)

You are trying to fit a model and are given $p=30$ predictor variables to choose from. Ultimately, you want your model to be interpretable, so you decide to use Best Subset Selection.

How many different models will you end up considering?:

Answer: 2^{30}

EXPLANATION

Each predictor can either be included or not included in the model. That means that for each of the 30 variables there are two options. Thus, there are 2^{30} potential models.

Note: Don't ever try to fit that many models! It is too many and that is why Best Subset Selection is rarely used in practice for say $p=10$ or larger.

[Hide Answer](#)

You have used 1 of 5 submissions

6.3.R2 (1/1 point)

How many would you fit using Forward Selection?:

Answer: $1+30*(30+1)/2$

EXPLANATION

For Forward Selection, you fit $(p-k)$ models for each $k=0, \dots, p-1$. The expression for the total number of models fit is on pg 15 of the notes: $p(p+1)/2+1$.

Hide Answer

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